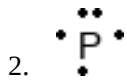


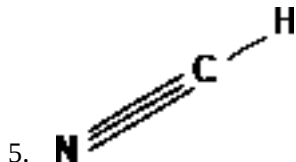
CHM111 Practice Exam on chapter(s): 9, 10

1. Answer: X belongs to Group 2 $\therefore$ X is Mg-like and has 2 valence electrons.



3. Electron-group arrangement: $\ll$ Tetrahedral $\gg$; Shape: $\ll$ Tetrahedral $\gg$; Ideal bond angle(s): $\ll$ 109.5° $\gg$

4. BaCl_2 $\therefore$ larger ion(s) $\rightarrow$ weaker electrostatic attraction; see Coulomb's law ($E \propto \frac{q_1 q_2}{r}$)



NOTE: this figure does not show any possible lone pairs and lone electrons. To get credit, YOU MUST show all existing lone pairs and lone electrons.

6. $+1 \therefore FC = \text{valence } e^- - \text{unshared } e^- - \frac{1}{2} \text{shared } e^- \therefore FC = 3 - 0 - \frac{1}{2}(4)$
7. Au, Cu, Zn, Ga $\therefore$ metallicity is highest in the bottom-left of the periodic table
8. covalent $\therefore$ H_2O is a covalent compound.
9. The following compounds have a dipole moment: OF_2